

Bootstrap — Topic 1: Introduction

Study paper format: notes (Roman Urdu + English technical terms) + active recall Q&A at the end.

1. Bootstrap kya hai?

Bootstrap ek **free, open-source CSS framework** hai jo responsive aur mobile-first websites jaldi banane mein madad karta hai.

Har button, navbar, form, card ya grid ke liye CSS khud se likhne ki bajaye, Bootstrap ready-made CSS classes provide karta hai. Isay ek **toolbox** samjho jismein UI components pehle se bane hue hain.

Bina Bootstrap:

html

```
<button class="my-btn">Click Me</button>
```

css

```
.my-btn{
  background:#0d6efd;
  color:white;
  padding:12px 24px;
  border:none;
  border-radius:6px;
}
```

Bootstrap ke sath:

html

```
<button class="btn btn-primary">Click Me</button>
```

Koi custom CSS ki zaroorat nahi.

Bootstrap do cheezon se mil kar banta hai

1. CSS — styles deta hai:

- Buttons, Forms, Tables, Cards, Alerts, Grid System, Typography, Utilities

2. JavaScript — kuch components ko JS chahiye hoti hai:

- Modal, Dropdown, Carousel, Tooltip, Collapse, Offcanvas

Bina JS ke: dropdown open nahi hoga, modal show nahi hoga.

Mobile-First Approach

Bootstrap **mobile-first** design se bana hai — pehle chhote screens ke liye design hota hai, phir tablets aur desktops ke liye scale up hota hai.

```
html
```

```
<div class="col-12 col-md-6 col-lg-4">
```

- Mobile → full width
- Tablet → half width
- Desktop → one-third width

(Grid System chapter mein detail se aayega.)

Bootstrap kyun bana?

Bootstrap se pehle:

- Har developer CSS scratch se likhta tha
- Alag alag browsers websites ko differently render karte thay

- Responsive websites banana bohat mehnat talab tha

Bootstrap ne solve kiya:

- Consistent styling
- Responsive layouts
- Ready-made components

2. Bootstrap ke Advantages

#	Advantage	Point
1	Fast Development	CSS likhne ki bajaye sirf classes use karo (e.g. <code>p-3 m-4 bg-danger rounded</code>)
2	Responsive	Phones, tablets, laptops, large screens sab par auto-adapt
3	Consistent Design	Buttons, forms, cards sab ka look similar → professional appearance
4	Huge Component Library	Navbar, Cards, Alerts, Forms, Pagination, Badges, Toasts, Accordion, Carousel, Modal — sab ready
5	Easy to Learn	HTML + basic CSS pata ho to sirf classes seekhni parti hain (e.g. <code>text-danger fw-bold fs-2</code>)
6	Excellent Documentation	Har component ke sath code, preview, explanation
7	Cross Browser Support	Chrome, Firefox, Edge, Safari sab par kaam karta hai

3. Bootstrap ke Disadvantages

#	Disadvantage	Point
1	Websites similar lag sakti hain	Beginners default styles hi use karte hain
2	Extra CSS	Thousands of styles hote hain, aksar 20% hi use hota hai — baqi bhi download hota hai
3	Class names yaad rakhna	CSS likhne ki bajaye class names remember karne parte hain (mt-5, mb-3, px-4, text-center)
4	Limited Custom Design	Unique design chahiye to phir bhi custom CSS likhni parti hai
5	JavaScript Dependency	Modal, Collapse, Carousel — bina JS kaam nahi karte

4. Bootstrap CDN

CDN = Content Delivery Network. Isse Bootstrap ko **bina download kiye** internet se directly use kar sakte hain.

CSS CDN:

```
html

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.min.css"
rel="stylesheet">
```

JS CDN (`</body>` se pehle place karte hain):

```
html
```

```
<script  
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.bundle.mi  
</script>
```

Complete Example:

```
html  
  
<!DOCTYPE html>  
<html>  
<head>  
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-scale=1.0">  
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/css/bootstrap.m  
</head>  
<body>  
<button class="btn btn-primary">Hello Bootstrap</button>  
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.8/dist/js/bootstrap.b  
</body>  
</html>
```

CDN kab use karein: Learning, small projects, practice, simple websites, CodePen/JSFiddle.

CDN kab NA use karein: Large professional projects — wahan npm se local install karte hain taake assets bundle, file size optimize, customization, aur version management ho sake.

5. Bootstrap Installation with npm

Vite / Webpack / React jaise tools ke sath npm preferred method hai.

Step 1 — Project banao (Vite example):

```
bash  
npm create vite@latest
```

Step 2 — Bootstrap install karo:

```
bash  
npm install bootstrap
```

Step 3 — CSS import karo ((`main.js`) mein):

```
javascript  
import 'bootstrap/dist/css/bootstrap.min.css';
```

Step 4 — JS import karo:

```
javascript  
import 'bootstrap/dist/js/bootstrap.bundle.min.js';
```

Ab Bootstrap project mein available hai.

6. Bootstrap Folder Structure

```
node_modules/  
  bootstrap/  
    dist/  
      css/  
        bootstrap.css  
        bootstrap.min.css  
      js/  
        bootstrap.js  
        bootstrap.bundle.js  
        bootstrap.min.js  
        bootstrap.bundle.min.js  
    scss/  
    icons/
```

- **dist/css** — compiled CSS jo browser use karta hai. **bootstrap.css** = readable version, **bootstrap.min.css** = production ke liye minified.
- **dist/js** — JavaScript files (JS bundle wali detail neeche).
- **scss** — Bootstrap ki source SCSS files, variables edit karke customize kar sakte hain (beginners ko zaroorat nahi).

7. Bootstrap JavaScript Bundle

Do JS versions hote hain:

1. **bootstrap.js** — sirf Bootstrap JS. Kuch components ko positioning ke liye **Popper.js** chahiye hoti hai (dropdowns, tooltips), jo separately include karni parti hai.
2. **bootstrap.bundle.js** — Bootstrap JS + Popper.js dono ek file mein.

Recommended: hamesha **bootstrap.bundle.min.js** use karo, kyunke Popper already included hoti hai.

JS-dependent components: Modal, Dropdown, Tooltip, Popover, Collapse, Accordion, Carousel, Offcanvas.

JS-independent (CSS-only): Buttons, cards, typography, spacing, colors, grid system.

Summary Table

Topic	Key Point
Bootstrap	Responsive, mobile-first websites ke liye CSS framework
Advantages	Fast dev, responsive, reusable components, consistency, easy to learn
Disadvantages	Similar-looking sites, extra CSS, class memorization, limited custom design, JS dependency
CDN	Learning aur small projects ke liye fastest way
npm Installation	React, Vite, Webpack, production projects ke liye best
Folder Structure	<code>dist/css</code> → CSS, <code>dist/js</code> → JS, <code>scss</code> → source files
JS Bundle	<code>bootstrap.bundle.min.js</code> = Bootstrap JS + Popper (recommended)

Practice Task

CDN se Bootstrap connect karke ek HTML page banao jismein:

- Heading: **"Welcome to Bootstrap"**
- Blue button (`btn-primary`) — **"Get Started"**
- Green button (`btn-success`) — **"Learn More"**
- Ek responsive `container` jo content ko center kare aur top margin (`mt-5`) ho

Active Recall — Bootstrap Introduction

| Khud se answer karne ki koshish karo, phir neeche scroll karke check karo.

Q1. Bootstrap kya hai, aur ye kis approach par bana hai?

Q2. Bootstrap kin do cheezon se mil kar banta hai?

Q3. Mobile-first approach ka matlab kya hai? (`col-12 col-md-6 col-lg-4`) ka example de kar samjhao.

Q4. Bootstrap ke bina developers ko konse 3 problems face karni parti thin?

Q5. Fast Development advantage ko ek CSS vs Bootstrap class example se explain karo.

Q6. Bootstrap ke 3 disadvantages likho.

Q7. CDN kya hota hai, aur ye kab use karna chahiye vs kab nahi?

Q8. npm se Bootstrap install karne ke 4 steps kya hain?

Q9. (`dist/css`), (`dist/js`), aur (`scss`) folders mein kya farq hai?

Q10. (`bootstrap.js`) aur (`bootstrap.bundle.js`) mein kya farq hai? Konsa use karna chahiye aur kyun?

Q11. Konse components ko JavaScript chahiye hoti hai, aur konse sirf CSS se kaam kar jate hain?

Q12. (`bootstrap.min.css`) aur (`bootstrap.css`) mein kya difference hai?

Answers

<details> <summary>Answers dekhne ke liye click karein</summary>

A1. Bootstrap ek free, open-source CSS framework hai jo responsive aur mobile-first websites jaldi build karne mein help karta hai. Ye mobile-first approach par bana hai — pehle small screens design hote hain, phir tablets aur desktops ke liye scale up hote hain.

A2. CSS (styles: buttons, forms, cards, grid, etc.) aur JavaScript (interactive components jaise modal, dropdown, carousel).

A3. Mobile-first ka matlab hai pehle chhote screen ke liye design karna, phir bara karna. `col-12 col-md-6 col-lg-4` mein mobile par full width, tablet par half width, aur desktop par one-third width milti hai.

A4. (1) Har developer CSS scratch se likhta tha, (2) alag browsers websites ko differently render karte thay, (3) responsive websites banana bohat mehnat talab tha.

A5. Bina Bootstrap: `padding:15px; margin:20px; background:red; border-radius:5px;` custom CSS likhni parti hai. Bootstrap ke sath sirf class use hoti hai: `<div class="p-3 m-4 bg-danger rounded">`.

A6. Koi bhi 3: websites similar lag sakti hain, extra unused CSS download hoti hai, class names yaad rakhni parti hain, limited custom design ke liye phir bhi CSS likhni parti hai, JS dependency (Modal, Collapse, Carousel).

A7. CDN (Content Delivery Network) se Bootstrap ko bina download kiye internet se directly link kar sakte hain. Learning, small projects, practice, aur simple websites ke liye use karo. Large professional projects mein npm se local install karna better hai (bundling, optimization, customization, version control ke liye).

A8. (1) `npm create vite@latest` se project banao, (2) `npm install bootstrap` se install karo, (3) `main.js` mein `import 'bootstrap/dist/css/bootstrap.min.css';` se CSS import karo, (4) `import 'bootstrap/dist/js/bootstrap.bundle.min.js';` se JS import karo.

A9. `dist/css` mein compiled CSS files hoti hain jo browser use karta hai. `dist/js` mein JavaScript files hoti hain. `scss` mein Bootstrap ki source SCSS files hoti hain jinhe customize karke compile kar sakte hain.

A10. `bootstrap.js` mein sirf Bootstrap JS hoti hai — Popper.js separately include karni parti hai. `bootstrap.bundle.js` mein Bootstrap JS + Popper.js dono included hoti hain. Almost hamesha `bootstrap.bundle.min.js` use karna chahiye kyunke Popper already saath mein hoti hai.

A11. JS chahiye: Modal, Dropdown, Tooltip, Popover, Collapse, Accordion, Carousel, Offcanvas. Sirf CSS se kaam ho jata hai: Buttons, cards, typography, spacing, colors, grid system.

A12. `bootstrap.css` readable/unminified version hai (development ke liye), jabke `bootstrap.min.css` minified version hai jo production ke liye use hoti hai (smaller file size, faster load).

</details>